Anchor Selection for Node Localization Using Deep Reinforcement Learning

A Final Project End Semester Report

submitted by

SAI PRANAV MADUPU (CS22B1027)

in partial fulfilment of requirements
for the award of the degree of

BACHELOR OF TECHNOLOGY

Department of Computer Science and Engineering
INDIAN INSTITUTE OF INFORMATION TECHNOLOGY,
DESIGN AND MANUFACTURING, KANCHEEPURAM

May 2026

DECLARATION OF ORIGINALITY

I, **Sai Pranav Madupu**, with Roll No: **CS22B1027** hereby declare that the material presented in the Project Report titled **Anchor Selection for Node Localization Using Deep Reinforcement Learning** represents original work carried out by me in the **Department of Computer Science and Engineering** at the Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram.

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Sai Pranav Madupu

Place: Chennai

Date: 11.05.2026

CERTIFICATE

This is to certify that the report titled **Anchor Selection for Node Localization Using Deep Reinforcement Learning**, submitted by **Sai Pranav Madupu (CS22B1027)**, to the Indian Institute of Information Technology, Design and Manufacturing Kancheepuram, in partial fulfilment of requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** is a bonafide record of the work done by him under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Dr. Preeth R
Project Internal Guide
Assistant Professor
Department of Computer Science and Engineering
IIITDM Kancheepuram, Chennai - 600 127

Place: Chennai

Date:

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to **Dr. Preeth R** for his invaluable guidance and constant encouragement. His expertise and mentorship have been instrumental in shaping the work titled **"Anchor Selection for Node Localization Using Deep Reinforcement Learning"** and in guiding its ongoing development and refinement.

I am equally grateful to **Ms. Rajathi S (Ph.D. Scholar, CS23D0006)** for her continuous support and technical guidance that greatly enhanced my understanding of this project.

I extend my heartfelt thanks to the **Department of Computer Science and Engineering, IIITDM Kancheepuram**, for providing the academic environment and resources necessary to carry out this work.

I would also like to thank all my professors and mentors whose teaching, encouragement, and academic insights have inspired me to explore the domains of Internet of Things and Artificial Intelligence in practical research applications.

ABSTRACT

Accurate node localization in Ultra-Wideband(UWB) networks plays an important role in many of the Internet of Things and indoor positioning applications. But, due to it's continuous utilization of a selected optimal anchors increases energy consumption, which leads to reduce in overall network lifetime. Efficient anchor selection becomes critical in achieving a balance between localization accuracy and energy consumption.

This project deals with the problem of anchor selection in Ultra-Wideband (UWB) based indoor localization systems. Although UWB technology provides very high positioning accuracy, using multiple anchors continuously increases the energy consumption and reduces the overall lifetime of the network. Therefore, finding a proper balance between localization accuracy and energy efficiency is an important challenge that needs to be addressed. In this work, the anchor selection problem is modeled as a sequential decision making task and a simulation framework is developed which includes Two-Way Ranging (TWR) for distance measurement, stochastic LoS and NLoS channel modeling, and an IEEE 802.15.4a compliant energy consumption model.

Several baseline anchor selection methods, including Random selection, Nearest Neighbor selection, GDOP-based selection, and a DQLEL-inspired approach, are implemented and evaluated under a unified simulation framework to ensure fair comparison. This setup allows us to analyze the trade-off between localization accuracy and the energy efficiency across different anchor selection strategies. Advanced learning methods such as Deep Reinforcement Learning, Meta-Reinforcement Learning and RL^2 are further explored within the same environment to study their adaptability and generalization capabilities under varying indoor conditions.

KEYWORDS: Ultra-Wideband (UWB); Node Localization; Anchor Selection; Deep Reinforcement Learning; Deep Q-Network (DQN); Internet of Things (IoT).

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ABBREVIATIONS

UWB	Ulra Wide Band
LoS	Line of Sight
NLoS	Non-Line of Sight
ToF	Time of Flight
TDoA	Time Difference of Arrival
GDOP	Geometric Dilution of Precision
DQN	Deep Q-Network
CNN	Convolutional Neural Network
ECDF	Empirical Cumulative Distribution Function
GPS	Global Positioning System

NOTATION

(x, y)	True position of the mobile tag
$(\hat{x}, \hat{y})$	Estimated position of the mobile tag
d_i	True distance between tag and i^{th} anchor
$d_i^{measured}$	Measured distance using TWR
N	Total number of anchor nodes
$\mathcal{A}_t$	Selected anchor subset at time step t
E_t	Localization error at time step t
η_i	Gaussian ranging noise component
ϵ	Exploration rate in ϵ-greedy policy

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

In today's world, IoT devices are being used in almost every field. As these devices are being deployed in large numbers, there is an increasing need for systems that can track the location of objects or people inside buildings. Outdoor location tracking is usually done using GPS, but GPS signals do not reach properly inside buildings because of walls and other obstructions. Therefore, for indoor environments, alternative localization approaches are required.

UWB technology is one such approach that has shown very good results for indoor positioning. It works by sending very short pulses of radio signals over a wide range of frequencies. Because of this, it can measure distances very precisely and provide location estimates with centimeter-level accuracy. In a UWB localization system, fixed nodes called anchors are placed at known positions in the environment. A mobile device, referred to as a tag, estimates its position by measuring its distance from each anchor.

This type of system is useful in many practical applications. For example, in environments such as coal mines or bunkers, it is important to continuously track the position of gas sensors to detect hazardous situations like carbon monoxide leakage at an early stage. UWB localization is also widely used in warehouses for asset tracking, and in industrial environments for guiding autonomous guided vehicles (AGVs).

However, continuously using all available anchors for localization leads to increased computational overhead and higher energy consumption. Since anchor nodes operate on limited battery power, excessive usage reduces the operational lifetime of the system. On the other hand, selecting fewer anchors can reduce localization accuracy due to poor geometric coverage. This creates a trade-off between energy efficiency and positioning accuracy.

Another challenge is that most of the existing strategies do not adapt to changing environments. Therefore there is a need for an intelligent anchor selection technique that can adapt to changing environmental conditions and create a trade-off between the loc

1.2 UWB-Based Node Localization

The UWB based positioning works by transmitting very short radio pulses over the whole frequency band. In a system of UWB based localization system, there are multiple anchor nodes deployed at fixed and known location inside the indoor environment. The distance of the mobile tag from anchors are determined through ranging estimation techniques such as ToA and TWR. After getting the estimated distance of the mobile tag from anchors, localization algorithm like trilateration or multilateration can be used to calculate the position of the mobile tag. The estimated position depends on many variables such as measurement error, geometry distribution of anchors, number of anchors, LoS, NLoS and so on. Therefore, a suitable subset of anchors should be selected at each time step.

1.3 Objectives of the work

In this project, we propose a energy-efficient and accurate anchor selection strategy for UWB-based indoor localization systems. The entire system is simulated using multiple anchor nodes at fixed locations in a specified indoor environment and the position of a mobile tag is calculated using UWB-based ranging technique. The proposed anchor selection strategy uses a Deep Reinforcement Learning (DRL) framework to select an appropriate subset of anchors at each time instant, while balancing between localization accuracy and energy efficiency.

The contributions of this work are:

- Indoor UWB localization environment modeling for simulation.
- DRL-based framework for anchor selection design.

- Anchor selection algorithm implementation and integration with the localization system.
- Performance of system evaluating using error, and consumed energy on 10 unseen environments.

CHAPTER 2

LITERATURE REVIEW

2.1 Overview of UWB Localization Techniques

Ultra-Wideband (UWB) technology has become very popular for indoor localization because of its wide bandwidth, good time resolution, and ability to handle multipath signals effectively. Compared to narrowband systems, UWB signals can separate closely arriving signal components, which allows very accurate distance measurements up to centimeter level. Mazhar et al. [2] provide a detailed study on this and show that UWB is well suited for precise indoor localization. Time based techniques like Time of Arrival (TOA) and Time Difference of Arrival (TDOA) are commonly used in UWB systems because they make good use of the precise timing properties of UWB signals.

Table 2.1: Comparison of Indoor Localization Technologies (Hajiakhondi et al. [1])

Technology	Advantages	Limitations
UWB	Gives very high localization accuracy and can handle multipath signals effectively.	Has limited coverage range and the deployment cost is relatively high.
BLE	Consumes very low power and provides moderate coverage range.	Positioning accuracy is limited and the system is sensitive to noise and interference.
RFID	Requires very less energy and active tags can cover a reasonably large area.	Can only provide coarse level localization and is not suitable for precise positioning.
WiFi	Can make use of already existing network infrastructure without additional hardware.	Needs complex signal processing and is highly affected by multipath interference.
Zigbee	Supports low power consumption and can be scaled easily for large sensor networks.	Has a limited data rate and the communication range is relatively short.

The DQLEL framework presented by Hajiakhondi et al. [1] uses a 2-D TDOA based localization model where the position of a mobile user is estimated using time differences between synchronized UWB beacons. The authors model the wireless channel using Nakagami-m fading and test the system under both Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) conditions. Their results show that selecting the right subset of beacons can improve localization accuracy and also reduce energy consumption at the same time.

In another work, Nosrati et al. [3] propose a Deep Reinforcement Learning (DRL) based method for selecting anchor points intelligently using a mobile UWB sensor. The selected anchors are then used along with a CNN based model to estimate the 2-D position of the target. This method achieves a Mean Absolute Error (MAE) of around 0.09 m, which is better than fixed and random anchor selection methods.

A detailed review of traditional techniques such as RSSI, TOA, TDOA, and AOA is provided by Mazhar et al. [2], where it is shown that time based methods are more suitable for UWB systems. However, these methods are still sensitive to synchronization errors and NLoS conditions, which makes intelligent anchor selection an important research area.

2.2 Anchor Selection Techniques

Anchor selection is a very important step in improving localization accuracy and reducing unnecessary computation. Early methods used geometric metrics like Mean Squared Error (MSE) and Geometric Dilution of Precision (GDOP) to select the best anchors, as discussed by Nosrati et al. [3]. These methods group anchors into clusters and select those that give the least estimation error.

Albaidhani et al. [4] proposed an anchor selection method for UWB based indoor positioning where anchor nodes are grouped and their geometric quality is evaluated using Weighted GDOP (WGDOP) along with Linearized Least Squares (LLS) estimation. The method selects the anchor subset that gives the least positioning error in indoor environments. In a related work, Albaidhani and Alsudani [5] further studied GDOP based anchor selection and showed that choosing geometrically favorable anchor combinations can improve localization accuracy. Even though both of these methods improve positioning performance, they mainly depend on geometric metrics and do not consider long-term energy balancing among the anchor nodes.

In the DQLEL framework proposed by Hajiakhondi et al. [1], the anchor selection problem is modeled as a Markov Decision Process (MDP). Here, the mobile user acts as an agent and selects two UWB beacons based on its current location and the remaining battery level of the anchors. The reward function considers both localization error and battery usage, which allows the system to optimize both objectives together. The reward is defined as:

$$R(s_t, a_t) = \begin{cases} \frac{1}{MD_t \cdot CR_t}, & \text{if } MD_t \leq MD_{th},\ CR_t \leq CR_{th} \\ -MD_t \cdot CR_t, & \text{otherwise} \end{cases}$$

where MD_t denotes the mean localization error at time step t, and CR_t represents the energy consumption rate (or cost) associated with the selected anchors. The thresholds MD_{th} and CR_{th} are predefined limits for acceptable localization error and energy consumption, respectively. This reward formulation encourages the agent to select anchors that achieve low error and low energy usage, while penalizing poor selections that violate these constraints.

The DQLEL framework by Hajiakhondi et al. [1] shows better convergence and more balanced energy usage when compared with methods like Random Node Selection (RNS), Nearest Neighbor Node Selection (NN-NS), and other DRL methods that do not consider energy.

In the work by Nosrati et al. [3], the anchor placement problem is also modeled as an MDP and solved using Proximal Policy Optimization (PPO). In this case, a mobile UWB sensor learns the best locations to place anchors by maximizing the received signal power and reducing multipath effects. This work shows that reinforcement learning is becoming increasingly useful for adaptive anchor configuration.

2.3 Reinforcement Learning in Wireless Networks

Reinforcement Learning (RL) has gained substantial popularity for solving dynamic resource allocation and decision making problems in wireless networks. The problem is framed as a Markov Decision Process (MDP), in which the agent performs an action in an environment in order to maximize cumulative future rewards.

In the DQLEL framework proposed by Hajiakhondi et al. [1], Deep Q-Learning is employed to select the best UWB anchors for localization. The states consist of the user position and battery levels of anchors. All possible combinations of anchors form the action space. A Convolutional Neural Network (CNN) approximates the Q-function, thereby providing a scalable solution for large state and action spaces. The system converges within 350 epochs and yields reduced localization errors as compared to baseline methods.

In the work by Nosrati et al. [3], Proximal Policy Optimization (PPO), an actor-critic DRL algorithm, is applied to derive robust anchor placement strategies. The UWB mobile sensor perceives signal strength and position to decide upon the movement it takes next. PPO appears stable and is efficient at exploring complex indoor environments.

Both of these works prove that RL based frameworks are adaptable to varying channel conditions, multipath effects, and energy constraints which are characteristics of an intelligent localization system.

2.4 Generalization and Advanced RL Techniques (Meta-RL, Domain Generalization, RL)

Many recent developments in reinforcement learning (RL) have demonstrated impressive ability of models to generalize well in various environments using methods like domain randomization. The study conducted by Tobin et al. [6] proposed to train models on a vast number of randomized simulation environments to be capable of learning task-relevant features that generalize irrespective of the environment in question.

Rather than training the model on a single static setting, the models are exposed to variations in the system parameters, signal noise conditions, spatial configurations etc. This technique is highly relevant in the case of UWB localization systems because the environments vary widely from one setting to another. Environmental characteristics like beacon positions, multipath propagation and Line-of-Sight (LoS) or Non-Line-of-Sight (NLoS) conditions change significantly across the locations. The present work draws inspiration from this work to include multi-environment training so as to learn a better beacon selection policy that is robust in diverse environments.

Meta-learning further enhances adaptivity by enabling models to quickly adapt to new tasks or environments with minimal additional data. A famous example of meta-learning is the Model-Agnostic Meta-Learning (MAML) framework introduced by Finn et al. [7], wherein models are trained to achieve a favorable parameter initialization which could later be quickly fine-tuned in a few steps from just a small amount of data for a new task. In the context of reinforcement learning, the agent becomes adaptable to new environments with merely a few interactions. This is particularly important for indoor localization systems as the environment can be dynamic, with varying factors impacting the system. By incorporating meta-learning, the present work aims to develop a robust beacon selection policy capable of generalizing to previously unseen UWB environments.

Expanding further on meta-learning principles, the RL^2 framework proposed by Duan et al. [8] used recurrent neural networks-based policy capable of implicit adaptation without having to update model parameters each time the environment is changed. The model used an LSTM network which encodes historical observations, actions, and rewards, thereby achieving an inherent adaptive mechanism without any explicit retraining. This is highly useful for environments that are partially observable and dynamic, such as in UWB localization system, where factors like noise and varying signal conditions may affect the system. Inspired by the work, the present work explores the application of LSTM-based RL for robust beacon selection across various previously unseen environments.

2.5 Baseline Methods for Comparison

To analyze the performance of different anchor selection strategies, baseline methods were implemented using the simulation setup described in section 3.3. The experimental configuration follows the IEEE 802.15.4a UWB energy model and incorporates LoS/NLoS channel modeling assumptions. Using this unified setup, the performance of conventional anchor selection strategies as well as the DQLEL framework was reproduced and evaluated under identical conditions.

The following baseline methods were considered:

- **Random Anchor Selection:** Anchors are selected randomly at each time step without considering geometry or energy constraints.
- **Nearest Anchor Selection:** The anchors with minimum estimated distance to the tag are selected at each step.
- **GDOP-based Anchor Selection:** Anchors are selected by minimizing the Geometric Dilution of Precision (GDOP) metric to improve geometric positioning quality.
- **DQLEL (Deep Q-Learning Energy-Optimized LoS/NLoS):** A reinforcement learning-based anchor selection strategy inspired by the DQLEL framework, which jointly considers localization accuracy and battery deviation in its reward formulation.

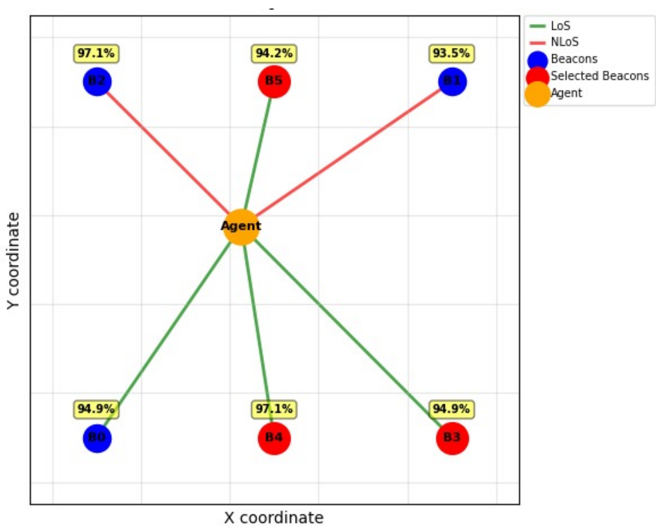

Figure 2.1: Illustration of Anchor Selection in the Simulated Environment. Green links represent LoS connections, while red links indicate NLoS conditions. The agent dynamically selects anchors based on link quality and signal conditions.

2.6 Research Gaps Identified

Even though many good methods have been proposed, there are still some important gaps in the existing research on UWB based localization.

- Most traditional anchor selection methods use fixed geometric metrics like GDOP and MSE. These methods cannot adapt to dynamic LoS and NLoS changes that occur in real indoor environments.
- Many existing UWB localization systems depend on strict time synchronization methods like TDOA and TOA, which increase the complexity and cost of the infrastructure, especially in large scale deployments. Two-Way Ranging (TWR) does not need tight global synchronization among anchors, which makes it more practical and scalable for real world use.
- Although TWR has slightly higher ranging error compared to TOA and TDOA based methods, this error can be reduced by using intelligent anchor selection and learning based optimization techniques.
- Many DRL based methods focus mainly on improving localization accuracy and do not give enough attention to long-term energy sustainability and network lifetime.

From the above discussion, it is clear that there is a strong need for intelligent, energy aware, and adaptive anchor selection methods that can jointly improve localization accuracy, handle NLoS conditions, and extend network lifetime. The proposed work in this project aims to contribute in this direction by using Deep Reinforcement Learning for energy efficient UWB anchor selection in dynamic indoor environments.

CHAPTER 3

SYSTEM MODEL AND PROBLEM FORMULATION

3.1 System Model

The system considered in this work consists of a set of fixed Ultra-Wideband (UWB) anchor nodes deployed in a two-dimensional indoor environment and a mobile UWB tag whose position needs to be estimated. Let N denote the total number of anchor nodes placed at known coordinates $\{(x_i, y_i)\}_{i=1}^{N}$. The mobile tag is located at an unknown position (x, y) within the coverage area of these anchors.

Two-Way Ranging (TWR) is used to measure the distances between the tag and the selected anchors. Unlike Time Difference of Arrival (TDoA) based systems, TWR does not require strict global time synchronization among anchors, making it more suitable for scalable indoor deployments.

At each localization step, only a subset of anchors is activated for ranging instead of using all available anchors. The choice of this subset directly influences localization accuracy, energy consumption, and overall network lifetime.

3.2 Problem Formulation

The anchor selection task is modeled as a sequential decision-making problem. Using all N anchors at every time step increases energy consumption and computational overhead. Conversely, selecting too few anchors may degrade localization accuracy due to poor geometric configuration.

Let $\mathcal{A}_t \subseteq \{1, 2, \ldots, N\}$ denote the subset of anchors selected at time step t. The objective is to determine a selection strategy that achieves a balance between:

- Minimizing localization error,

- Maintaining balanced energy consumption among anchor nodes,
- Extending overall network lifetime.

3.3 Simulation Setup

The simulation environment is designed to represent a realistic indoor deployment scenario similar to the configuration adopted in the DQLEL framework [1]. A square-shaped indoor area is considered, where N UWB anchor nodes are uniformly deployed along the boundaries. The mobile tag moves randomly within the area during each episode to emulate practical mobility patterns.

3.3.1 Channel and Propagation Model

To capture realistic wireless behavior, both Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) propagation conditions are modeled. Similar to [1], the wireless channel is assumed to experience fading and measurement noise. The ranging error is modeled as:

$$d_i^{measured} = d_i^{true} + \eta_i$$

where η_i is modeled as zero-mean Gaussian noise with variance dependent on the propagation condition.

For LoS links, the noise variance is relatively small, representing stable ranging conditions. For NLoS links, an additional positive bias component is introduced to reflect excess delay caused by multipath propagation and signal obstruction. This models the practical scenario where NLoS conditions introduce larger ranging errors compared to LoS links.

3.3.2 Energy Consumption Model

The energy consumption model follows the IEEE 802.15.4a compliant UWB transceiver model, similar to the formulation used in the DQLEL framework [1]. This model accounts for energy consumed during transmission, reception, signal processing, and circuit operations.

Each anchor node is initialized with a finite battery level $B_i^{(0)}$. During each TWR interaction, energy is consumed based on transmit, receive, and processing operations.

The battery update for anchor i at time step t is given by:

$$B_i^{(t+1)} = B_i^{(t)} - E_{tx} - E_{rx} - E_{proc}$$

For implementation simplicity, these components are aggregated into a single energy cost per interaction ΔE:

$$B_i^{(t+1)} = B_i^{(t)} - \Delta E$$

The network lifetime is defined as the duration until any anchor battery falls below a predefined operational threshold.

The parameters used for the IEEE 802.15.4a UWB energy model are summarized in Table 3.1.

Table 3.1: Key Parameter Settings for IEEE 802.15.4a UWB Transceiver Model

Symbol	**Value**	**Symbol**	**Value**
L_{SP}	1024	P_{SYN}	30.6
L_{PHR}	16	P_{ADC}	2.2
N_p	$\{1, 3, 5, \ldots, 15\}$	P_{GEN}	2.8
E_p	4.5	P_{LNA}	9.4
R_{base}	1	P_{EST}	10.08
P_{COR}	10.08	P_{VGA}	22

CHAPTER 4

PROPOSED METHODOLOGY

4.1 Overview of Proposed Framework

In a UWB-based indoor localization system, multiple anchor nodes are deployed at fixed positions within the environment. At any given time step, only a selected subset of these anchors is used to estimate the position of the mobile tag. The primary objective of the anchor selection process is to identify the most suitable combination of anchors such that the position of the tag can be estimated with good accuracy, while also ensuring that the battery levels of all anchor nodes are consumed in a balanced manner over time. It is important to note that this is not a one-time decision. The system must repeatedly make this selection at every time step, and each selection decision has a direct impact on the remaining battery levels of the anchors, which in turn affects future decisions.

In order to address this problem, a Reinforcement Learning (RL) based approach is proposed in this work. The core idea behind this approach is straightforward. Rather than relying on a fixed set of rules or geometric criteria to select anchors, the system is trained to learn good selection strategies through interaction with the environment. During training, the system tries out different anchor combinations, observes how accurately the localization was performed as a result, and gradually improves its decision-making ability over time. This learning process is conceptually similar to how a person improves at a task through repeated practice and experience.

The framework proposed in this work uses two reinforcement learning techniques called Meta-Reinforcement Learning (Meta-RL) and RL^2. These two techniques are selected because they help the system not only learn how to pick the right anchors in a given environment, but also quickly change its strategy when the environment is different. At each step, the system looks at the current situation, chooses a set of anchors, finds the position of the tag, and gets a score that tells it how well it did in terms of both location accuracy and how evenly the battery was used across all anchors. By going through this process many times during training, the system slowly learns to make decisions that keep both accuracy and energy usage in a good balance.u

4.2 Meta-Reinforcement Learning for Anchor Selection

Conventional reinforcement learning approaches are typically trained for one specific environment. When the environment changes, the performance of such a system tends to degrade significantly because the learned policy was optimized only for the original training conditions. In practical indoor environments, conditions do not remain constant. Anchor nodes may be placed at different positions, walls and obstacles may block or reflect signals, and the nature of the propagation path between the tag and the anchors may vary from one location to another. Due to all of these factors, a system that has been trained for only one fixed setting cannot be expected to consistently deliver good results across different deployments.

To address this limitation, Meta-Reinforcement Learning (Meta-To overcome this limitation, Meta-Reinforcement Learning (Meta-RL) is used in the proposed framework. In essence, the main concept of Meta-RL is that the system is trained on a large variety of environments such that it learns to adapt to a new environment to which it is deployed for the first time. One can consider a simple analogy of student practicing various kinds of problems in order to be ready for any type of new problem that the teacher may give. The system trained on many different environments, on the other hand, can handle the new environment even though it has never experienced this specific environment.

In the context of the present work, each distinct environment configuration, such as a different room layout, a different arrangement of anchor nodes, or a different set of channel conditions, is treated as a separate task. The system is trained sequentially across many such tasks during the training phase. As a result of this multi-task training, when the system is evaluated in a new environment that was not part of the training set, it is still able to perform reasonably well because it has already learned a general strategy for adapting to new situations.

This property of Meta-RL is particularly valuable for indoor localization applications, where the operating environment is highly unpredictable and can change in ways that are difficult to anticipate or model in advance.

4.3 RL^2 (LSTM-Based Policy Learning)

RL^2 is a Meta-RL method that makes use of a special type of recurrent neural network known as Long Short-Term Memory (LSTM) to enable the system to remember past interactions and use that information to make better decisions at each step. In a conventional reinforcement learning system, the decision at each time step is based solely on the current observed state of the environment. In contrast, RL^2 allows the system to also take into account the history of past interactions, such as which anchor subsets were selected in previous steps, how much battery was consumed by each anchor, and how accurate the localization results were. By incorporating this historical information into the decision-making process, the system is able to make more informed and effective anchor selection decisions.

In the proposed framework, the LSTM network is for keeping track of important temporal information such as how the battery levels of the anchor nodes have been changing over successive time steps, how the wireless channel quality has been varying, and how the localization performance has been changing in recent steps. All of this historical information is encoded and stored within the LSTM in the form of a hidden state vector, which is updated automatically at every time step as new observations are received.

At each time step, the system receives the current state of the environment as input. This includes information such as the estimated current position of the tag, the remaining battery level of each anchor node, and the localization error recorded in the most recent step. This current state information is combined with the hidden state maintained by the LSTM, and the network then produces a policy that determines which subset of anchors should be selected for the current time step.

The primary advantage of using RL^2 in this framework is that the system does not need to be explicitly retrained every time the environment changes. Instead, it is capable of adapting its anchor selection strategy on its own, based on the recent history of observations and outcomes that it has accumulated through interaction with the new environment.

4.4 Training Strategy and Learning Framework

This section describes the training setup and implementation details of the proposed reinforcement learning based UWB beacon selection framework. The system is designed to evaluate and compare four different learning strategies, namely standard Deep Q-Network (DQN), Domain Generalization DQN, Meta-Reinforcement Learning (Meta-RL), and RL^2 with LSTM. Each of these strategies represents a different approach to the anchor selection problem, and the goal is to study how well each one performs in terms of localization accuracy and energy efficiency.

4.4.1 Overview of Learning Approaches

The proposed framework implements four distinct learning strategies for adaptive beacon selection. A brief description of each approach is given below.

- **Standard DQN:** In this approach, the agent is trained on a single fixed environment. A Multi-Layer Perceptron (MLP) is used as the function approximator to learn a fixed policy for anchor selection.
- **Domain Generalization DQN:** In this approach, the agent is trained across multiple environments with varying characteristics. The goal is to learn a policy that can generalize well to new environments without requiring any additional training.
- **Meta-RL (MAML):** This approach trains the agent to find a good parameter initialization such that the model can quickly adapt to a new environment with only a small number of gradient update steps.
- **RL^2 (LSTM-based):** This approach uses a recurrent neural network architecture, specifically an LSTM, to allow the agent to implicitly learn the dynamics of the environment through the updates made to its hidden state over time.

In all four approaches, the objective is to select the best subset of 3 anchors out of a total of 6 available anchors at each time step. This results in a discrete action space of size 20.

4.4.2 State and Action Space

The state representation used by the agent varies depending on the learning approach being employed.

- **DQN and Domain Generalization DQN:** A 6-dimensional state vector is used, which consists of the normalized battery levels of all six anchor nodes.
- **Meta-RL:** A 15-dimensional state vector is used, which includes the battery levels of all anchors, distance estimates between the tag and each anchor, and LoS or NLoS indicators for each anchor link.
- **RL^2:** A 6-dimensional state vector consisting of the battery levels is used as the input at each time step. The temporal dependencies in the system are captured implicitly through the hidden state of the LSTM network.

The total number of possible actions, which corresponds to the number of ways in which 3 anchors can be selected from a set of 6, is given by the following combination formula:

$$|\mathcal{A}| = \binom{6}{3} = 20$$

4.4.3 Network Architecture

For the DQN-based approaches, a three-layer fully connected neural network is used as the Q-network. The architecture is as follows:

- Input layer: accepts the state vector and passes it to a layer of 64 neurons
- Hidden layer: 64 neurons connected to another layer of 64 neurons
- Output layer: 64 neurons connected to 20 output units, each corresponding to a Q-value for one possible anchor subset

For the RL^2 approach, an LSTM-based architecture is used instead. The details of this architecture are as follows:

- An LSTM layer with a hidden state size of 128
- A fully connected layer with 128 input units and 128 output units
- An output layer with 128 input units and 20 output units corresponding to the Q-values for each possible action

4.4.4 Training Configuration

All models are implemented using the PyTorch deep learning framework and are trained on a system equipped with a GPU to speed up the training process. The key hyperparameters used during training are listed below.

- Optimizer: Adam
- Learning rate: 0.001
- Discount factor (γ): 0.99
- Replay buffer size: $10{,}000$ transitions
- Mini-batch size: 32
- Target network update frequency: every 1000 training steps

An ϵ-greedy exploration strategy is employed throughout the training process. The value of ϵ is initially set to 1.0 and is gradually reduced to a minimum value of 0.01 as training progresses. This allows the agent to explore widely in the early stages of training and to gradually shift towards exploiting the policy it has learned as training continues.

4.4.5 Reward Function

A multi-objective reward function is used to guide the learning process. This reward function is designed to simultaneously encourage good localization accuracy, favorable anchor geometry, and balanced energy consumption. The reward at each time step is computed as follows:

$$R = 0.6 \cdot G - 0.3 \cdot B - 0.1 \cdot L - 0.1 \cdot D$$

where each term is defined as follows:

- G: A measure of the geometric quality of the selected anchor configuration
- B: A penalty term for imbalance in the battery levels of the anchor nodes
- L: A penalty term for the presence of NLoS links among the selected anchors
- D: A penalty term based on the Geometric Dilution of Precision (GDOP) of the selected anchor subset

This reward formulation encourages the agent to select anchor subsets that are geometrically well-distributed, have sufficient remaining battery, and maintain reliable Line-of-Sight links wherever possible.

4.4.6 Domain Generalization Training

For the Domain Generalization approach, the agent is trained across multiple simulated environments, each having different characteristics. The objective is to learn a single policy that performs well across all environments without requiring environment-specific tuning.

The key parameters that are varied across these environments are:

- Grid size: varies from 10×10 to 30×30
- Anchor positions: randomized for each environment
- Probability of LoS conditions: varies from 0.4 to 0.8
- Initial battery levels and battery consumption rates: randomized for each environment

A shared policy network is trained jointly across all these environments. During training, the agent interacts with different environments in each iteration, allowing it to learn a generalized strategy that is not specific to any single configuration. This improves the robustness of the learned policy and enables it to perform well in previously unseen scenarios.

Algorithm 1 Domain Generalization Training for Anchor Selection (Tobin et al. [6])

Input: Set of environments $\{\mathcal{E}_1, \mathcal{E}_2, ..., \mathcal{E}_K\}$, policy parameters θ
Output: Generalized policy π_θ
for each training iteration **do**
    for each environment $\mathcal{E}_k$ **do**
        Initialize environment $\mathcal{E}_k$
        Observe initial state s_0
        for each time step t **do**
            Select action $a_t \sim \pi_\theta(a|s_t)$
            Execute action and observe reward r_t, next state s_{t+1}
            Store transition (s_t, a_t, r_t, s_{t+1})
        end for
    end for
    Update policy parameters θ using data from all environments
end for

4.4.7 Meta-Reinforcement Learning (MAML)

To improve adaptability to new environments, a Meta-Reinforcement Learning approach based on Model-Agnostic Meta-Learning (MAML) is considered. Unlike standard RL, which learns a single policy, MAML aims to learn model parameters that can be quickly adapted to new tasks with minimal training.

In this context, each environment configuration is treated as a separate task. The training process consists of two levels of optimization:

- **Inner loop:** The model is adapted to a specific environment using a small number of gradient updates.
- **Outer loop:** The meta-parameters are updated based on performance across multiple tasks.

This allows the model to learn how to learn, enabling faster adaptation to new and unseen environments.

4.4.8 RL^2 with LSTM

In the RL^2 approach, the policy is represented using a recurrent neural network, specifically an LSTM. Instead of explicitly adapting the model parameters for each new environment, the adaptation is handled internally through the hidden state of the LSTM.

Algorithm 2 MAML-Based Meta-Reinforcement Learning for Anchor Selection (Finn et al. [7])

Input: Task distribution $\{\mathcal{T}_i\}$, initial parameters θ
Output: Meta-trained parameters θ
for each meta-training iteration **do**
    Sample batch of tasks $\{\mathcal{T}_i\}$
    for each task $\mathcal{T}_i$ **do**
        Collect trajectories using policy π_θ
        Inner Update:

$$\theta_i' = \theta - \alpha \nabla_\theta \mathcal{L}_{\mathcal{T}_i}(\theta)$$

        Collect trajectories using adapted parameters θ_i'
    end for
    Outer Update:

$$\theta \leftarrow \theta - \beta \sum_i \nabla_\theta \mathcal{L}_{\mathcal{T}_i}(\theta_i')$$

end for

At each time step, the model takes as input the current state along with information from the previous step (such as previous action and reward). The LSTM updates its hidden state, which acts as a memory of past interactions.

The hidden state is reset at the beginning of each episode and evolves over time as the agent interacts with the environment. This allows the agent to implicitly learn the characteristics of the current environment and adjust its behavior accordingly.

4.4.9 Training and Evaluation Setup

The models are trained on a collection of diverse simulated environments and are subsequently evaluated on a separate set of unseen test environments in order to assess their generalization capability. The overall training and evaluation setup involves the following:

- Training is performed across multiple randomized environments with varying characteristics
- Evaluation is performed on a separate set of environments that were not used during training
- Performance is assessed using two main metrics, namely the localization error and the network lifetime of the anchor nodes

Algorithm 3 RL^2 - LSTM-Based Policy Learning (Duan et al. [8])

Input: Set of environments $\{\mathcal{E}_i\}$, LSTM parameters θ
Output: Trained policy π_θ
for each episode **do**
    Sample environment $\mathcal{E}_i$
    Initialize hidden state $h_0 = 0$
    Observe initial state s_0
    for each time step t **do**
        Form input (s_t, a_{t-1}, r_{t-1})
        Update hidden state:

$$h_t = \text{LSTM}(h_{t-1}, input_t)$$

        Select action:

$$a_t \sim \pi_\theta(a|h_t)$$

        Execute action and observe reward r_t, next state s_{t+1}
    end for
    Update parameters θ using collected trajectories
end for

This training and evaluation setup is designed to ensure that the policies learned by each approach are robust and can be applied effectively in real-world indoor localization scenarios.

CHAPTER 5

RESULTS AND ANALYSIS

5.1 Overview

The reinforcement learning and meta-learning based anchor selection methods proposed in this work were tested and evaluated under two different types of simulated indoor environments:

1. **Randomized Environments:** In this setup, the size of the environment, the positions of the beacon nodes, the probability of Line-of-Sight conditions, and the noise characteristics were all generated randomly within certain predefined ranges. This was done to check how well each method performs when the deployment conditions are highly varied and unpredictable.
2. **Normally Distributed Environments:** In this setup, the dimensions of the environment were generated using a Gaussian distribution with a mean value of $\mu = 20$ and a standard deviation of $\sigma = 5$. This creates indoor layouts that are more statistically balanced while still having enough variety to test the generalization ability of the methods.

All models were first trained on a set of different simulated environments and then tested on new environments that were never used during training. This was done to measure how accurately each method can find the position of the tag and how well it can adapt to environments it has not seen before.

5.2 Evaluation Metrics

The following metrics were used to evaluate and compare the performance of all the methods.

5.2.1 Localization Accuracy Metrics

- **Mean Localization Error:** This is the average straight-line distance between the actual position of the tag and the position estimated by the system. A lower value means better accuracy.
- **RMSE (Root Mean Square Error):** This metric measures the overall localization accuracy while giving more weight to larger errors. A lower RMSE indicates better and more consistent performance.
- **Standard Deviation:** This measures how consistent the localization results are. A lower standard deviation means the system gives more stable and predictable results.
- **P95 Error:** These represent the localization error values below which 95% of all test cases fall, respectively. These metrics are useful for understanding how the system behaves in the worst cases.
- **ECDF (Empirical Cumulative Distribution Function):** This provides a complete picture of how the localization errors are distributed across all test cases, from the smallest to the largest.

5.2.2 Network Lifetime Metrics

- **First Node Death (FND):** This is the time step at which the first anchor node completely runs out of battery and stops working.
- **Half Node Death (HND):** This is the time step at which half of all the anchor nodes in the network have run out of battery.
- **Coverage Loss:** This is the time step at which fewer than three anchor nodes are still working, meaning the system can no longer perform reliable localization.
- **Energy Threshold:** This is the time step at which any one of the anchor nodes drops below a minimum battery level that has been set in advance.

5.3 Environment Distribution Analysis

Figures 5.1, 5.2, and 5.2 show the distributions of the environments that were generated for training and evaluation purposes.

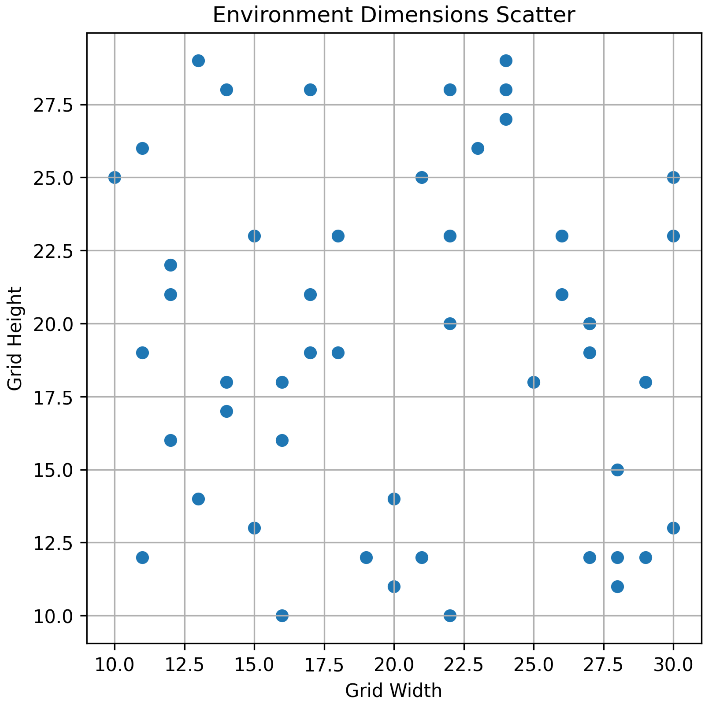

Figure 5.1: Scatter Plot of Generated Environment Dimensions

The scatter plot shows that the generated environments cover a wide range of different sizes and spatial configurations, which helps in testing the robustness of the methods under diverse conditions.

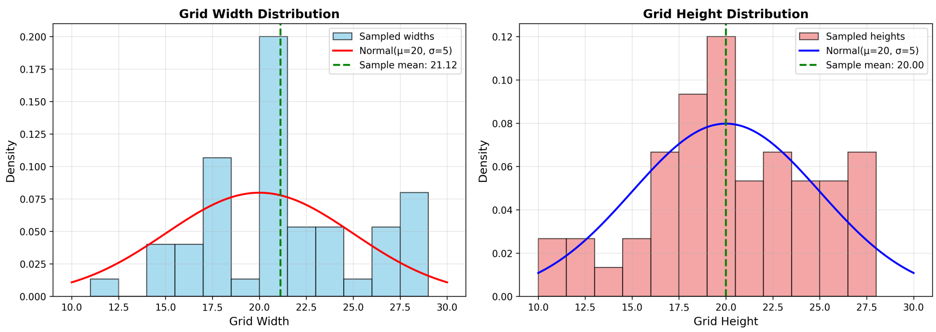

Figure 5.2: Width and Height Distributions with Gaussian Fit

The histogram plots show that the generated environment dimensions follow a Gaussian distribution centered around a mean of $\mu = 20$ with a standard deviation of $\sigma = 5$. This setup helps in creating indoor layouts that are realistic and statistically balanced, while still having enough variation to properly test the generalization capability of the proposed methods.

5.4 Localization Performance Analysis

5.4.1 Performance on Randomized Environments

Table 5.1: Localization Error Statistics on Randomized Environments

Method	Mean(m)	RMSE(m)	Std Dev(m)	P95(m)
Random	0.177	0.202	0.120	0.480
Nearest Neighbor	0.273	0.249	0.210	0.780
GDOP	0.147	0.189	0.110	0.420
DQN (Avg 40)	0.134	0.158	0.095	0.298
Domain Gen DQN	0.178	0.167	0.120	0.312
Meta RL	**0.112**	**0.153**	0.090	0.333
RL^2 LSTM	0.127	0.157	**0.085**	**0.255**

The results obtained from the randomized environment tests are summarized in Table 5.1. It can be seen that Meta-RL achieved the lowest mean and median localization error among all the methods, which shows that it is the most accurate and best at adapting to environments with highly varied and unpredictable conditions. RL^2 LSTM achieved the lowest standard deviation and the lowest 95th percentile error, which means that its results were the most consistent and it produced fewer very large errors compared to the other methods.

On the other hand, traditional methods such as Random selection and Nearest Neighbor selection showed much higher localization errors and greater variation in results, especially in environments where the anchor geometry was poor or where NLoS conditions were present.

5.4.2 Performance on Normally Distributed Environments

Table 5.2: Localization Error Statistics on Normally Distributed Environments

Method	Mean(m)	RMSE(m)	Std Dev(m)	P95(m)
Random	0.149	0.316	0.277	0.843
Nearest Neighbor	0.161	0.301	0.252	0.688
GDOP	0.194	0.236	0.211	0.580
DQN (Avg 40)	0.173	0.312	0.256	0.638
Domain Gen DQN	0.267	0.449	0.360	0.944
Meta RL	0.128	**0.179**	**0.124**	**0.354**
RL^2 LSTM	0.157	0.246	0.189	0.494

The results from the normally distributed environment tests are shown in Table 5.2. Meta-RL once again gave the best overall performance by achieving the lowest RMSE, standard deviation, and 95th percentile error among all the methods tested. RL^2 LSTM also performed well, with lower variance compared to the standard DQN approach.

The Domain Generalization DQN method gave the worst results in this set of experiments, with the highest localization error across almost all metrics. This suggests that learning a single fixed policy to handle a wide variety of environments, without any form of explicit adaptation, is a very difficult task and does not work well in practice.

When compared to the randomized environment results, the normally distributed environments produced more stable and consistent localization behavior overall. This is likely because the statistically balanced nature of these environments leads to more predictable signal conditions during testing.

5.5 ECDF Analysis

The Empirical Cumulative Distribution Function (ECDF) was used to get a better understanding of how the localization errors are distributed across all test cases for each method. In an ECDF plot, a curve that rises more steeply towards the left side of the graph indicates that a larger proportion of test cases have low localization error, which means the method is performing better and more consistently.

5.5.1 ECDF Analysis on Randomized Environments

Figure 5.3 shows the ECDF comparison for all methods on the randomized environment setup.

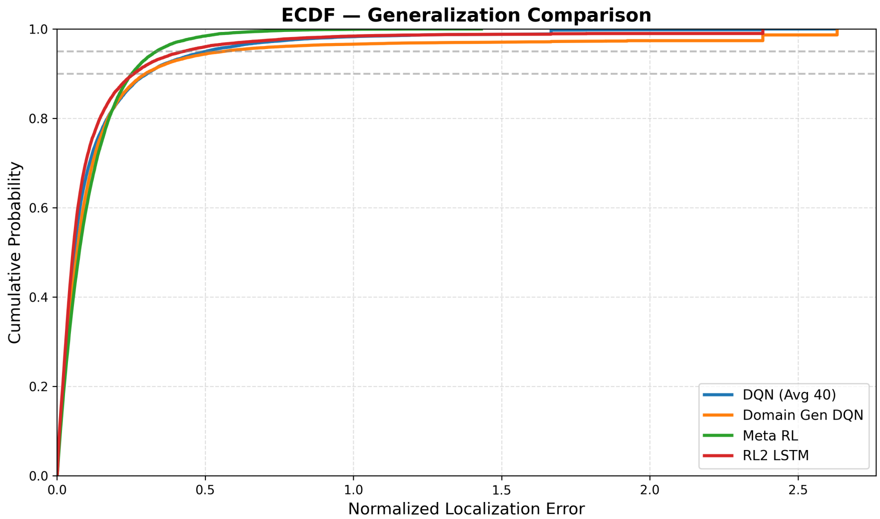

Figure 5.3: ECDF Comparison on Randomized Environments (in meters)

From the figure, it can be observed that Meta-RL and RL^2 LSTM achieve a steeper rise in the low-error region compared to all other methods. This confirms that these two approaches produce a higher proportion of low-error localization results, even in environments with highly varied and unpredictable conditions.

The traditional methods, namely Random selection and Nearest Neighbor selection, show a slower rise in the ECDF curve, which means they produce a larger number of high-error cases. The Domain Generalization DQN also performs poorly in this setup, further confirming the difficulty of generalizing a single policy to highly diverse environments.

5.5.2 ECDF Analysis on Normally Distributed Environments

Figure 5.4 shows the ECDF comparison for all methods on the normally distributed environment setup.

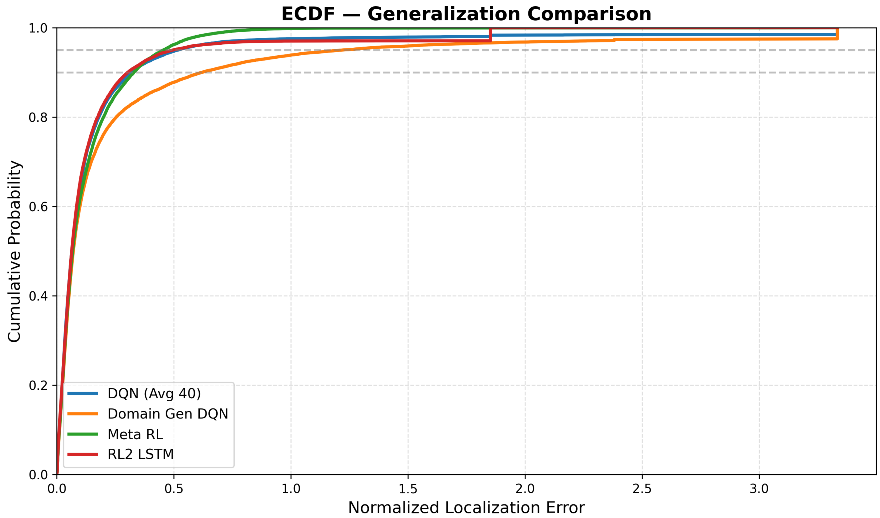

Figure 5.4: ECDF Comparison on Normally Distributed Environments (in meters)

Compared to the randomized environment results, the ECDF curves for the normally distributed environments are generally smoother and show fewer extreme error cases across all methods. Among all the methods, Meta-RL shows the steepest cumulative error curve, which confirms that it gives the most robust and consistent localization results in these environments as well.

RL^2 LSTM also shows stable cumulative performance with very few cases of extremely large localization errors. The Domain Generalization DQN continues to show a slower rise and a wider spread of errors, which confirms that its ability to adapt to different environments is limited compared to the meta-learning based approaches.

Overall, the ECDF analysis confirms that meta-learning based approaches, particularly Meta-RL and RL^2 LSTM, provide better and more reliable localization performance compared to traditional methods and standard reinforcement learning approaches.

5.6 Network Lifetime Analysis

In addition to localization accuracy, the network lifetime performance of all the proposed anchor selection methods was also evaluated. Four metrics were used for this analysis, as described below.

- **First Node Death (FND):** The time step at which the first anchor node runs out of battery completely.
- **Half Node Death (HND):** The time step at which half of all anchor nodes have run out of battery.
- **Coverage Loss:** The time step at which fewer than three anchor nodes are still working, which means reliable localization is no longer possible.
- **Energy Threshold:** The time step at which any anchor node drops below a minimum predefined battery level.

The purpose of this analysis is to check whether the intelligent anchor selection methods proposed in this work also help in extending the overall operating life of the anchor network, in addition to improving localization accuracy.

5.6.1 Network Lifetime on Randomized Environments

The network lifetime results for the randomized environment setup are presented in Table 5.3.

Table 5.3: Network Lifetime Metrics on Randomized Environments (in time steps)

Method	**FND**	**HND**	**Coverage Loss**	**Threshold**
Random	246.5	262.5	278.5	223.5
Nearest Neighbor	252.5	277.5	281.5	226.5
GDOP	252.5	277.5	281.5	226.5
DQN	252.5	277.5	281.5	226.5
Domain Gen	252.5	277.5	281.5	226.5
Meta RL	252.5	277.5	281.5	226.5
RL^2 LSTM	252.5	277.5	281.5	226.5

From the table, it can be seen that the Random selection method shows slightly lower network lifetime values compared to all other methods, particularly in terms of the FND and HND metrics. All the remaining methods, whether heuristic-based or learning-based, show nearly identical network lifetime values across all four metrics.

This result suggests that the anchor selection strategies used in the current framework do not have a strong influence on how the battery of each anchor is consumed over time. Since most methods tend to spread the usage of anchors relatively evenly across the network, the overall energy consumption pattern remains similar for all methods, leading to very similar network lifetime values.

5.6.2 Network Lifetime on Normally Distributed Environments

The network lifetime results for the normally distributed environment setup are presented in Table 5.4.

Table 5.4: Network Lifetime Metrics on Normally Distributed Environments (in time steps)

Method	**FND**	**HND**	**Coverage Loss**	**Threshold**
Random	251.46	270.75	278.01	225.95
Nearest Neighbor	250.51	271.44	278.59	226.17
GDOP	250.51	271.44	278.59	226.17
DQN	250.51	271.44	278.59	226.17
Domain Gen	250.51	271.44	278.59	226.17
Meta RL	250.51	271.44	278.59	226.17
RL^2 LSTM	250.51	271.44	278.59	226.17

Compared to the randomized environment results, the normally distributed environments show slightly more stable network lifetime behavior across all methods. However, similar to the randomized setup, the differences among all the methods remain very small and are not significant.

These results confirm that the reinforcement learning and meta-learning based approaches proposed in this work are mainly focused on improving localization accuracy and generalization, and do not have a significant impact on energy consumption or network lifetime under the current reward function design. This is an important observation, as it suggests that if energy efficiency and longer network lifetime are desired goals, additional components such as battery balancing penalties or anchor usage regularization terms may need to be added to the reward function in future work.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This study examined and evaluated an intelligent anchor selection system for UWB indoor localization, utilizing reinforcement learning and meta-learning. The goal of this study was to enable a system to pinpoint a person or object in a building with increased precision while maintaining robust performance under various indoor settings including changes in anchor placement and signal blockage from walls between the tag and anchors. Four learning based approaches including Standard DQN, Domain Generalization DQN, Meta-RL, and RL^2 with LSTM, were examined in an indoor environment within a simulation. The simulation results show that each of the learning based approaches are superior to traditional selection methods like Random and Nearest Neighbor selection in more difficult indoor conditions.

The Meta-RL system showed superior performance in an unseen environment, yielding the minimum positional error. RL^2 always performed well across various environments with minimal fluctuation. The analysis of the error distributions also showed that the meta-learning systems had more consistent and smaller position errors which shows that these systems can handle challenging conditions well. The one weakness that was apparent in the current work is that in terms of energy conservation and the amount of time that anchor nodes were working, the energy consumption of the anchors did not greatly differ amongst the learning based systems and the traditional approaches, indicating room for further improvement in this system. The proposed system shows that reinforcement learning and meta-learning are very good approaches to creating effective indoor localization systems for the selection of anchors. It provided a strong foundation to the advancement of intelligence indoor positioning systems.

6.2 Future Work

The developed system provided excellent results but there are multiple aspects where this study can be improved and extended.

- **Real World Implementation:** All of the tests were run using a simulation of an indoor environment. A final step that needs to be taken is to test the presented algorithm using a UWB device and in a real world indoor environment to ascertain how well the system performs.
- **Improved Reward Function:** A reward function could be developed so that not only is the positioning accuracy better, but also that energy efficiency and system robustness are taken into account as well. This may lead to an even better method of learning how to position anchors.
- **Multimodal Localization:** This system relied only on UWB positioning. More accurate and more robust localization can be obtained by fusing UWB with another form of positioning such as IMU based or visual based positioning.
- **Better Generalization Algorithms:** In this paper we have presented Meta-RL and RL^2 that generalize well to a new environment, however more advanced meta-learning algorithms exist that could further increase the system's ability to generalize to unseen environments.

APPENDIX A

APPENDIX

A.1 Simulation Snapshots

This section presents additional simulation snapshots generated from the developed UWB localization environment. The figures illustrate different indoor scenarios with varying anchor layouts, LoS/NLoS link conditions, mobile tag positions, and selected anchor subsets during localization.

These visualizations help in understanding the behavior of the simulation environment under different configurations used during training and evaluation.

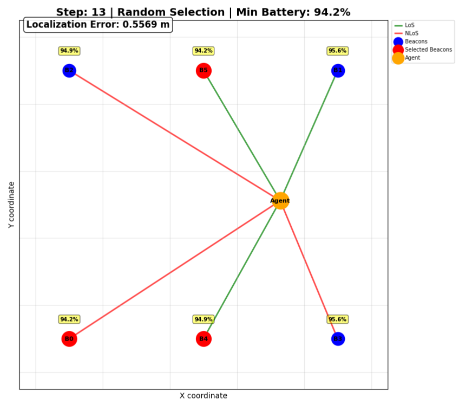

Figure A.1: Simulation Snapshot 1

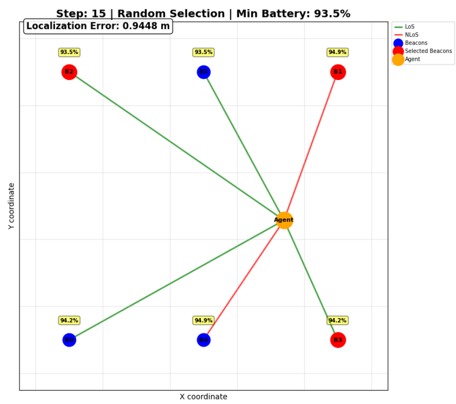

Figure A.2: Simulation Snapshot 2

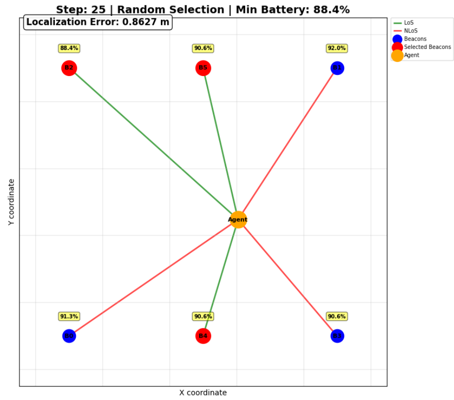

Figure A.3: Simulation Snapshot 3

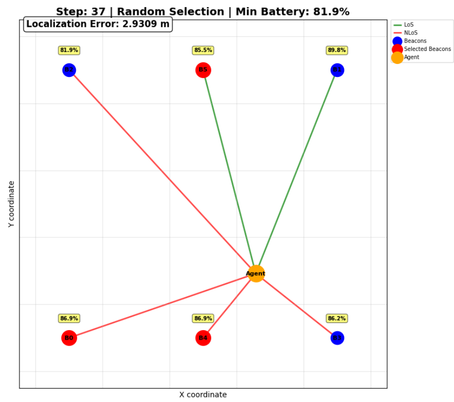

Figure A.4: Simulation Snapshot 4

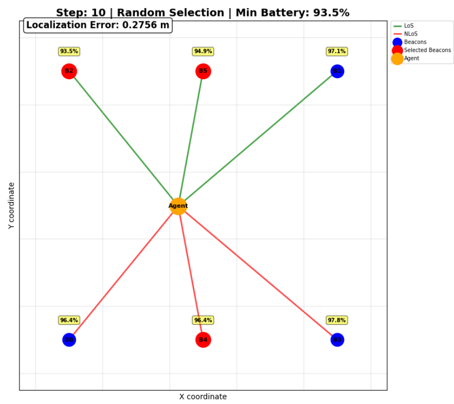

Figure A.5: Simulation Snapshot 5

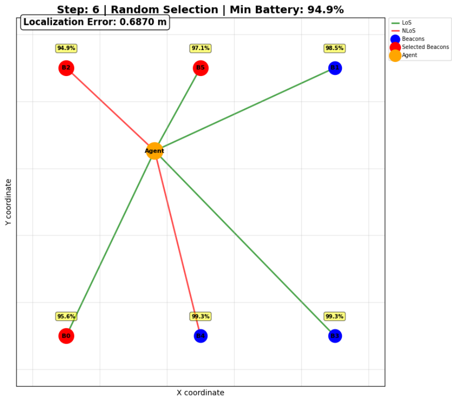

Figure A.6: Simulation Snapshot 6

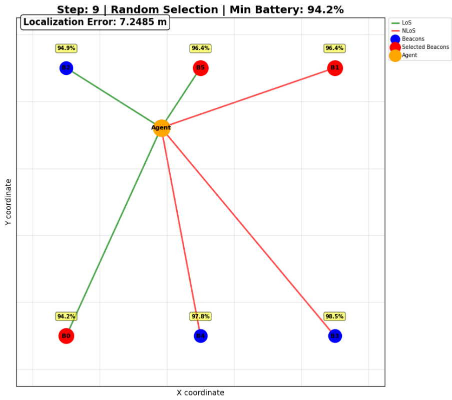

Figure A.7: Simulation Snapshot 7

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10	9/03/2026 - 15/03/2026	Analyzed mid-sem feedback & understood what kind of features the selection methods should have.	
11	16/03/2026 - 22/03/2026	Conducted literature review on Domain Generalization & Meta-RL.	
12	23/03/2026 - 29/03/2026	Conducted literature review on RL^2 LSTM.	
13	30/03/2026 - 05/04/2026	Implemented advanced CSR simulation.	
14	06/04/2026 - 12/04/2026	Implemented Domain Generalization & Meta RL approaches.	
15	13/04/2026 - 19/04/2026	Implemented RL^2 & Developed evaluation framework.	
16	20/04/2026 - 26/04/2026	Evaluated all the proposed methods & Developed Network Lifetime evaluation framework.	
17	27/04/2026 - 03/05/2026	Analyzed results & made changes to the evaluation framework.	
18	04/05/2026 - 10/05/2026	worked on project report & PPT.	
	End Semester Review Feedback		